

ALFAAL14460

## Iodine monobromide

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                      |                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br>Product Description: | <b>Iodine monobromide</b><br>Iodine monobromide                                                                                                                                                                                                                    |
| <b>Cat No. :</b>                     | <b>L14460</b>                                                                                                                                                                                                                                                      |
| <b>Synonyms</b>                      | Iodine Bromide.                                                                                                                                                                                                                                                    |
| <b>CAS No</b>                        | 7789-33-5                                                                                                                                                                                                                                                          |
| <b>Molecular Formula</b>             | Br I                                                                                                                                                                                                                                                               |
| <b>Supplier</b>                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| <b>Emergency Telephone Number</b>    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b>               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| <b>Uses advised against</b>          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                   |                                |                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------|
| <b>Physical State</b><br>Solid                                                                                                                    | <b>Appearance</b><br>Dark grey | <b>Odor</b><br>Odorless |
| <b>Emergency Overview</b><br>Causes severe skin burns and eye damage. May cause respiratory irritation. Sensitivity to light. Moisture sensitive. |                                |                         |

#### Classification of the substance or mixture

|                                                    |              |
|----------------------------------------------------|--------------|
| Skin Corrosion/Irritation                          | Category 1 B |
| Serious Eye Damage/Eye Irritation                  | Category 1   |
| Specific target organ toxicity - (single exposure) | Category 3   |

#### Label Elements



Signal Word

Danger

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**Hazard Statements**

H314 - Causes severe skin burns and eye damage

H335 - May cause respiratory irritation

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P310 - Immediately call a POISON CENTER or doctor

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Corrosive. Causes skin and eye burns. Causes serious eye damage. May cause respiratory irritation.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component            | CAS No    | Weight % |
|----------------------|-----------|----------|
| Iodine bromide (IBr) | 7789-33-5 | >95      |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

**Ingestion**

Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.

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**Most important symptoms and effects**

Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Self-Protection of the First Aider**

Use personal protective equipment as required. Avoid contact with skin, eyes or clothing.

**Notes to Physician**

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

The product causes burns of eyes, skin and mucous membranes.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing. Refer to protective measures listed in Sections 7 and 8

**Environmental Precautions**

Should not be released into the environment. Do not allow material to contaminate ground water system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Store under an inert atmosphere. Keep at temperatures between 2°C and 8 °C. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION**

## Iodine monobromide

## Control Parameters

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

## Exposure Controls

## Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** When using do not eat, drink or smoke. Remove and wash contaminated clothing and gloves, including the inside, before re-use.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

**Appearance** Dark grey  
**Physical State** Solid

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|                                         |                             |                                   |
|-----------------------------------------|-----------------------------|-----------------------------------|
| Odor                                    | Odorless                    |                                   |
| Odor Threshold                          | No data available           |                                   |
| pH                                      | No information available    |                                   |
| Melting Point/Range                     | 42 - 50 °C / 107.6 - 122 °F |                                   |
| Softening Point                         | No data available           |                                   |
| Boiling Point/Range                     | 116 °C / 240.8 °F           |                                   |
| Flash Point                             | No information available    | Method - No information available |
| Evaporation Rate                        | Not applicable              | Solid                             |
| Flammability (solid,gas)                | No information available    |                                   |
| Explosion Limits                        | No data available           |                                   |
| Vapor Pressure                          | No data available           |                                   |
| Vapor Density                           | Not applicable              | Solid                             |
| Specific Gravity / Density              | 4.410                       |                                   |
| Bulk Density                            | No data available           |                                   |
| Water Solubility                        | Soluble in water            |                                   |
| Solubility in other solvents            | No information available    |                                   |
| Partition Coefficient (n-octanol/water) |                             |                                   |
| Autoignition Temperature                | No data available           |                                   |
| Decomposition Temperature               | 116 °C                      |                                   |
| Viscosity                               | Not applicable              | Solid                             |
| Explosive Properties                    | No information available    |                                   |
| Oxidizing Properties                    | No information available    |                                   |
| Molecular Formula                       | Br I                        |                                   |
| Molecular Weight                        | 206.81                      |                                   |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|
| Stability                | Light sensitive. Moisture sensitive.                                                                                |
| Hazardous Reactions      | None under normal processing.                                                                                       |
| Hazardous Polymerization | No information available.                                                                                           |
| Conditions to Avoid      | Incompatible products. Excess heat. Exposure to air. Exposure to light. Exposure to moisture. Avoid dust formation. |
| Materials to avoid       | Alcohols. Strong oxidizing agents. Strong reducing agents. Strong acids. Strong bases.                              |

Hazardous Decomposition Products Hydrogen iodide. Bromine.

## SECTION 11. TOXICOLOGICAL INFORMATION

|                                        |                                                             |
|----------------------------------------|-------------------------------------------------------------|
| Product Information                    | No acute toxicity information is available for this product |
| (a) acute toxicity;                    |                                                             |
| (b) skin corrosion/irritation;         | Category 1 B                                                |
| (c) serious eye damage/irritation;     | Category 1                                                  |
| (d) respiratory or skin sensitization; |                                                             |
| Respiratory                            | No data available                                           |
| Skin                                   | No data available                                           |

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|                                            |                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------|
| (e) germ cell mutagenicity;                | No data available                                                                                |
| (f) carcinogenicity;                       | No data available<br>There are no known carcinogenic chemicals in this product                   |
| (g) reproductive toxicity;                 | No data available                                                                                |
| (h) STOT-single exposure;                  | Category 3                                                                                       |
| Results / Target organs                    | Respiratory system                                                                               |
| (i) STOT-repeated exposure;                | No data available                                                                                |
| Target Organs                              | No information available.                                                                        |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                                                          |
| Symptoms / effects, both acute and delayed | Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |

## SECTION 12. ECOLOGICAL INFORMATION

|                                 |                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects             | Do not empty into drains.                                                                                                                                 |
| Persistence and Degradability   |                                                                                                                                                           |
| Persistence                     | Soluble in water, Persistence is unlikely, based on information available.                                                                                |
| Degradability                   | Not relevant for inorganic substances.                                                                                                                    |
| Bioaccumulative Potential       | Bioaccumulation is unlikely                                                                                                                               |
| Mobility in soil                | The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils |
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors                                                                                 |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance                                                                                            |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance                                                                                            |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| Contaminated Packaging              | Do not reuse empty containers. Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                                                                                                               |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH                                                                                                                                                                                                                                                      |

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and harm aquatic organisms.

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

UN-No UN3260  
Proper Shipping Name Corrosive solid, acidic, inorganic, n.o.s.  
Technical Shipping Name Iodine bromide  
Hazard Class 8  
Packing Group II

IMDG/IMO

UN-No UN3260  
Proper Shipping Name Corrosive solid, acidic, inorganic, n.o.s.  
Technical Shipping Name Iodine bromide  
Hazard Class 8  
Packing Group II

IATA

UN-No UN3260  
Proper Shipping Name Corrosive solid, acidic, inorganic, n.o.s.  
Technical Shipping Name Iodine bromide  
Hazard Class 8  
Packing Group II

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component            | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|----------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Iodine bromide (IBr) | X                                                   | -                                       | X    | X     | 232-159-9 | X    | -   | X     | X    | X    | X    | KE-21024 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
Creation Date 08-Apr-2009  
Revision Date 16-May-2024  
Revision Summary New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and

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hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**